



Student Number	
Mark / 35	

Chemistry

Preliminary Course

Final Examination • 2005

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A Data Sheet and a Periodic Table are provided at the back of this paper and may be removed for convenience
- Write your Student Number at the top of this page

Total Marks – 35

Part A – 9 marks

- Attempt Questions 1 – 9
- Allow about 10 minutes for this part

Part B – 26 marks

- Attempt Questions 10 – 19
- Allow about 40 minutes for this part

Allow about 10 minutes for this part

A  B  C  D 

correct

Answer Box for Questions 1 – 9				
1	A ☐	B ☐	C ☐	D ☐
2	A ☐	B ☐	C ☐	D ☐
3	A ☐	B ☐	C ☐	D ☐
4	A ☐	B ☐	C ☐	D ☐
5	A ☐	B ☐	C ☐	D ☐
6	A ☐	B ☐	C ☐	D ☐
7	A ☐	B ☐	C ☐	D ☐
8	A ☐	B ☐	C ☐	D ☐
9	A ☐	B ☐	C ☐	D ☐

1 Study the reaction... $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$

Which of the statements cannot apply to this reaction?

- (A) One gram of H_2 reacts with one gram of Cl_2 to produce 2 grams of HCl .
- (B) One molecule of H_2 reacts with one molecule of Cl_2 to produce 2 molecules of HCl .
- (C) One litre of H_2 reacts with one litre of Cl_2 to produce 2 litres of HCl at constant conditions.
- (D) One mole of H_2 reacts with one mole of Cl_2 to produce 2 moles of HCl .

2 Which of the production sequences shows the extraction of copper from its ore to 99.9% pure copper?

- (A) froth flotation $\rightarrow$ crushing and grinding $\rightarrow$ smelting $\rightarrow$ electrolysis
- (B) crushing and grinding $\rightarrow$ froth flotation $\rightarrow$ smelting $\rightarrow$ electrolysis
- (C) crushing and grinding $\rightarrow$ smelting $\rightarrow$ froth flotation $\rightarrow$ electrolysis
- (D) smelting $\rightarrow$ crushing and grinding $\rightarrow$ froth flotation $\rightarrow$ electrolysis

3 Which of the following may **not** be a consequence of thermal pollution in water?

- (A) Reduction in dissolved oxygen.
- (B) Disruption of aquatic organisms breeding cycles.
- (C) Out of season migration of aquatic fauna.
- (D) Decrease in salt concentration.

4 At 100 kPa and 25°C , 4 litres of oxygen gas contain 1×10^{21} molecules.
Which of these gas samples also contains 1×10^{21} molecules under the same conditions?

- (A) 1 L of NH_3
- (B) 2 L of Cl_2
- (C) 4 L of CO_2
- (D) 8 L of He

5 To which reaction can Gay–Lussac’s law be applied?

- (A) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$
(B) $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$
(C) $2\text{H}_2\text{O}(\text{l}) + 2\text{Na}(\text{s}) \rightarrow 2\text{Na}^+(\text{aq}) + 2\text{OH}^-(\text{aq}) + \text{H}_2(\text{g})$
(D) $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$

6 Which property accounts for the moderate viscosity of water?

- (A) Specific heat capacity
(B) Hydrogen bonding
(C) Density
(D) Boiling point

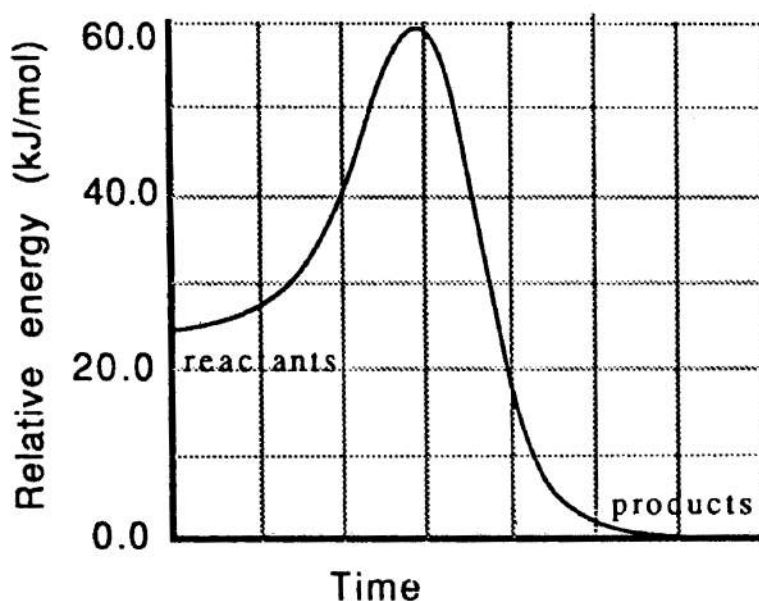
7 Metals X, M, Z and D are all very useful metals. The table shows selected properties of the metals...

<i>Property</i>	<i>X</i>	<i>M</i>	<i>Z</i>	<i>D</i>
<i>Ionisation energy (kJ mol⁻¹)</i>	584	751	896	766
<i>Percentage Abundance in Earth’s Crust</i>	8	0.07	0.00001	0.07

Use the data to list the metals according to increasing market price.

- (A) $X < M < D < Z$
(B) $D < X < Z < M$
(C) $X < Z < M < D$
(D) $Z < D < M < X$

Answer Questions 8 and 9 using this reaction pathway diagram...



- 8 Which statement correctly describes this reaction?
- (A) The reaction is endothermic and the surroundings will become cooler.
 - (B) The reaction is exothermic and the surroundings will become cooler.
 - (C) The reaction is endothermic and the surroundings will become warmer.
 - (D) The reaction is exothermic and the surroundings will become warmer.
- 9 Which statement is true if a catalyst is added to the system?
- (A) ΔH remains constant and the activation energy increases.
 - (B) Both ΔH and activation energy decrease.
 - (C) ΔH remains constant and the activation energy decreases.
 - (D) Both ΔH and activation energy remain constant.

Part B – 26 marks

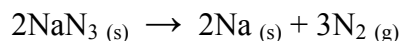
Attempt Questions 10 – 19

Allow about 40 minutes for this part

► *Show all relevant working in questions involving calculations.*

Question 10 (3 marks)

Air bags have saved thousands of lives and are now commonly fitted in new cars. An air bag inflates very rapidly by producing nitrogen gas from the decomposition of 100 grams of sodium azide...



- (a) Calculate the moles of sodium azide originally in the air bag. **(1 mark)**

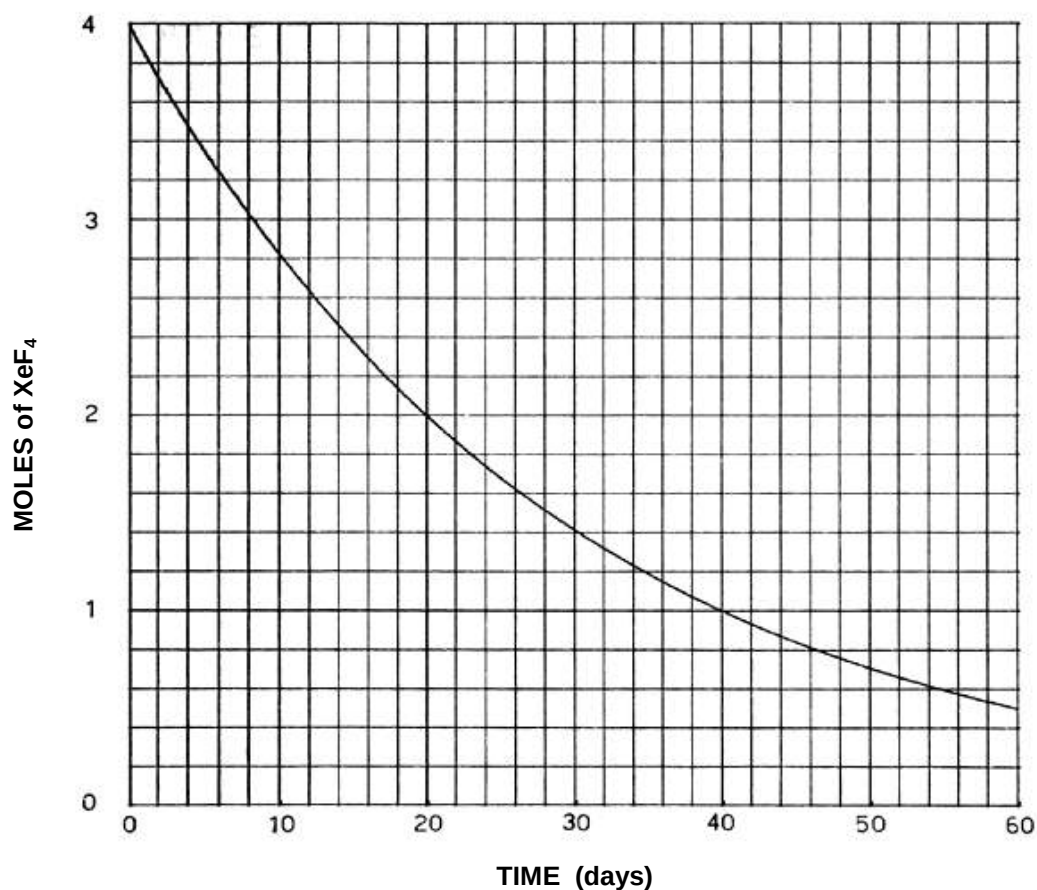
- (b) Calculate the moles of nitrogen produced and the resultant volume at 100 kPa and 25°C. **(2 marks)**

Question 11 (3 marks)

Xenon tetrafluoride is an unstable compound which self-decomposes...



The graph shows the decomposition of a pure, 4 mole sample of XeF_4 over a period of 60 days.



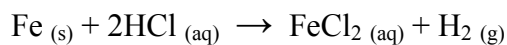
- (a) Calculate the number of moles of XeF_4 which has decomposed after 16 days. **(1 mark)**

- (b) On which day are there equal moles of reactant and products present? **(1 mark)**

- (c) Calculate the mass of F_2 present at 40 days. **(1 mark)**

Question 12 (2 marks)

Iron will react with hydrochloric acid and appear to “dissolve” ...

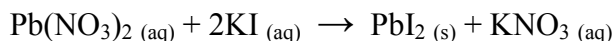


Ken Chemiski places a 2.51 g iron nail in a beaker and prepares some dilute, 1.00 mol L^{-1} HCl.

Calculate the volume of acid required to fully react with the nail.

Question 13 (3 marks)

Héloïse prepares some beautiful golden lead(II) iodide crystals by this precipitation reaction...



She reacts 25.0 mL of 0.100 mol L^{-1} potassium iodide with excess lead(II) nitrate solution.

(a) Calculate the moles of lead(II) nitrate which reacted. **(2 marks)**

(b) Calculate the mass of PbI_2 produced. **(1 mark)**

Question 14 (3 marks)

In your studies, you performed a practical investigation to observe the effect of temperature on reaction rate. Describe your experiment, including the observed results.

Question 15 (2 marks)

- (a) Identify one pollutant produced by the incomplete combustion of an organic compound. **(1 mark)**

- (b) Write a chemical equation to show the incomplete combustion of butane (C_4H_{10}). **(1 mark)**

Question 16 (2 marks)

The diagram shows liquid methanol, CH_3OH , being poured into a beaker of water.



Use the symbols, $\text{H}-\text{O}-\text{H}$ for water and $\begin{array}{c} \text{H} \\ | \\ \text{H}-\text{C}-\text{H} \\ | \\ \text{O}-\text{H} \end{array}$ for methanol to draw a diagram that illustrates the strongest intermolecular forces in the solution.

Question 17 (3 marks)

Outline the method required to prepare a 250 mL solution of 0.102 mol L^{-1} strontium chloride.

► *Write your answer in numbered sequential steps (1, 2, 3...).*

Question 18 (2 marks)

Tincture of iodine, a common antiseptic, is a 2% (w/w) solution of iodine in ethanol.

- (a) Calculate the mass of iodine crystals is required to prepare a 250 g sample of tincture of iodine? **(1 mark)**

- (b) Calculate the mass of ethanol is required? **(1 mark)**

Question 19 (3 marks)

In an experiment to measure the density of water at 28°C, an empty 100 mL measuring cylinder was found to have a mass of 150.5 grams. When 80.3 mL of water was poured into the cylinder, the mass of the cylinder and contents was 230.5 grams.

- (a) Calculate the density of water at 28°C based on the data. **(1 mark)**

- (b) When the measuring cylinder and water was sealed and kept inside a freezer overnight at – 10°C, the volume reading increased to 85.9 mL. Explain this observation. **(1 mark)**

- (c) When water freezes it expands. Why is this process important in Nature? **(1 mark)**

HIGHER SCHOOL CERTIFICATE EXAMINATION
Chemistry

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \quad \Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

KEY												KEY																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
Atomic Number	Symbol of element											Atomic Weight	Name of element																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
1 H 1.008 Hydrogen		5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.41 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton	37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.94 Molybdenum	43 Tc [97.91] Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon	55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57-71 Lanthanides	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.8 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po [209.0] Polonium	85 At [210.0] Astatine	86 Rn [222.0] Radon	87 Fr [223.0] Francium	88 Ra [226.0] Radium	89-103 Actinides	104 Rf [261.1] Rutherfordium	105 Db [262.1] Dubnium	106 Sg [266.1] Seaborgium	107 Bh [264.1] Bohrium	108 Hs [277] Hassium	109 Mt [268] Meitnerium	110 Ds [271] Darmstadtium	111 Rg [272] Roentgenium	112 Nh [284] Nihonium	113 Fl [289] Flerovium	114 Lv [293] Livermorium	115 Mc [288] Moscovium	116 Lr [260] Lawrencium	117 Ts [294] Tennessine	118 Og [294] Oganesson	119 Uue [289] Ununennium	120 Uuh [289] Unbinilium	121 Uut [288] Untrium	122 Uuq [287] Unquadrium	123 Uup [286] Unpentium	124 Uuq [285] Unsextium	125 Uuh [284] Unseptium	126 Uub [283] Unoctium	127 Uut [282] Unnonium	128 Uuq [281] Undecium	129 Uup [280] Undecium	130 Uub [279] Untrium	131 Uut [278] Untrium	132 Uuq [277] Unquadrium	133 Uup [276] Unpentium	134 Uub [275] Unsextium	135 Uut [274] Unseptium	136 Uub [273] Unoctium	137 Uut [272] Unnonium	138 Uuq [271] Undecium	139 Uup [270] Undecium	140 Uub [269] Untrium	141 Uut [268] Untrium	142 Uuq [267] Unquadrium	143 Uup [266] Unpentium	144 Uub [265] Unsextium	145 Uut [264] Unseptium	146 Uub [263] Unoctium	147 Uut [262] Unnonium	148 Uuq [261] Undecium	149 Uup [260] Undecium	150 Uub [259] Untrium	151 Uut [258] Untrium	152 Uuq [257] Unquadrium	153 Uup [256] Unpentium	154 Uub [255] Unsextium	155 Uut [254] Unseptium	156 Uub [253] Unoctium	157 Uut [252] Unnonium	158 Uuq [251] Undecium	159 Uup [250] Undecium	160 Uub [249] Untrium	161 Uut [248] Untrium	162 Uuq [247] Unquadrium	163 Uup [246] Unpentium	164 Uub [245] Unsextium	165 Uut [244] Unseptium	166 Uub [243] Unoctium	167 Uut [242] Unnonium	168 Uuq [241] Undecium	169 Uup [240] Undecium	170 Uub [239] Untrium	171 Uut [238] Untrium	172 Uuq [237] Unquadrium	173 Uup [236] Unpentium	174 Uub [235] Unsextium	175 Uut [234] Unseptium	176 Uub [233] Unoctium	177 Uut [232] Unnonium	178 Uuq [231] Undecium	179 Uup [230] Undecium	180 Uub [229] Untrium	181 Uut [228] Untrium	182 Uuq [227] Unquadrium	183 Uup [226] Unpentium	184 Uub [225] Unsextium	185 Uut [224] Unseptium	186 Uub [223] Unoctium	187 Uut [222] Unnonium	188 Uuq [221] Undecium	189 Uup [220] Undecium	190 Uub [219] Untrium	191 Uut [218] Untrium	192 Uuq [217] Unquadrium	193 Uup [216] Unpentium	194 Uub [215] Unsextium	195 Uut [214] Unseptium	196 Uub [213] Unoctium	197 Uut [212] Unnonium	198 Uuq [211] Undecium	199 Uup [210] Undecium	200 Uub [209] Untrium	201 Uut [208] Untrium	202 Uuq [207] Unquadrium	203 Uup [206] Unpentium	204 Uub [205] Unsextium	205 Uut [204] Unseptium	206 Uub [203] Unoctium	207 Uut [202] Unnonium	208 Uuq [201] Undecium	209 Uup [200] Undecium	210 Uub [199] Untrium	211 Uut [198] Untrium	212 Uuq [197] Unquadrium	213 Uup [196] Unpentium	214 Uub [195] Unsextium	215 Uut [194] Unseptium	216 Uub [193] Unoctium	217 Uut [192] Unnonium	218 Uuq [191] Undecium	219 Uup [190] Undecium	220 Uub [189] Untrium	221 Uut [188] Untrium	222 Uuq [187] Unquadrium	223 Uup [186] Unpentium	224 Uub [185] Unsextium	225 Uut [184] Unseptium	226 Uub [183] Unoctium	227 Uut [182] Unnonium	228 Uuq [181] Undecium	229 Uup [180] Undecium	230 Uub [179] Untrium	231 Uut [178] Untrium	232 Uuq [177] Unquadrium	233 Uup [176] Unpentium	234 Uub [175] Unsextium	235 Uut [174] Unseptium	236 Uub [173] Unoctium	237 Uut [172] Unnonium	238 Uuq [171] Undecium	239 Uup [170] Undecium	240 Uub [169] Untrium	241 Uut [168] Untrium	242 Uuq [167] Unquadrium	243 Uup [166] Unpentium	244 Uub [165] Unsextium	245 Uut [164] Unseptium	246 Uub [163] Unoctium	247 Uut [162] Unnonium	248 Uuq [161] Undecium	249 Uup [160] Undecium	250 Uub [159] Untrium	251 Uut [158] Untrium	252 Uuq [157] Unquadrium	253 Uup [156] Unpentium	254 Uub [155] Unsextium	255 Uut [154] Unseptium	256 Uub [153] Unoctium	257 Uut [152] Unnonium	258 Uuq [151] Undecium	259 Uup [150] Undecium	260 Uub [149] Untrium	261 Uut [148] Untrium	262 Uuq [147] Unquadrium	263 Uup [146] Unpentium	264 Uub [145] Unsextium	265 Uut [144] Unseptium	266 Uub [143] Unoctium	267 Uut [142] Unnonium	268 Uuq [141] Undecium	269 Uup [140] Undecium	270 Uub [139] Untrium	271 Uut [138] Untrium	272 Uuq [137] Unquadrium	273 Uup [136] Unpentium	274 Uub [135] Unsextium	275 Uut [134] Unseptium	276 Uub [133] Unoctium	277 Uut [132] Unnonium	278 Uuq [131] Undecium	279 Uup [130] Undecium	280 Uub [129] Untrium	281 Uut [128] Untrium	282 Uuq [127] Unquadrium	283 Uup [126] Unpentium	284 Uub [125] Unsextium	285 Uut [124] Unseptium	286 Uub [123] Unoctium	287 Uut [122] Unnonium	288 Uuq [121] Undecium	289 Uup [120] Undecium	290 Uub [119] Untrium	291 Uut [118] Untrium	292 Uuq [117] Unquadrium	293 Uup [116] Unpentium	294 Uub [115] Unsextium	295 Uut [114] Unseptium	296 Uub [113] Unoctium	297 Uut [112] Unnonium	298 Uuq [111] Undecium	299 Uup [110] Undecium	300 Uub [109] Untrium	301 Uut [108] Untrium	302 Uuq [107] Unquadrium	303 Uup [106] Unpentium	304 Uub [105] Unsextium	305 Uut [104] Unseptium	306 Uub [103] Unoctium	307 Uut [102] Unnonium	308 Uuq [101] Undecium	309 Uup [100] Undecium	310 Uub [99] Untrium	311 Uut [98] Untrium	312 Uuq [97] Unquadrium	313 Uup [96] Unpentium	314 Uub [95] Unsextium	315 Uut [94] Unseptium	316 Uub [93] Unoctium	317 Uut [92] Unnonium	318 Uuq [91] Undecium	319 Uup [90] Undecium	320 Uub [89] Untrium	321 Uut [88] Untrium	322 Uuq [87] Unquadrium	323 Uup [86] Unpentium	324 Uub [85] Unsextium	325 Uut [84] Unseptium	326 Uub [83] Unoctium	327 Uut [82] Unnonium	328 Uuq [81] Undecium	329 Uup [80] Undecium	330 Uub [79] Untrium	331 Uut [78] Untrium	332 Uuq [77] Unquadrium	333 Uup [76] Unpentium	334 Uub [75] Unsextium	335 Uut [74] Unseptium	336 Uub [73] Unoctium	337 Uut [72] Unnonium	338 Uuq [71] Undecium	339 Uup [70] Undecium	340 Uub [69] Untrium	341 Uut [68] Untrium	342 Uuq [67] Unquadrium	343 Uup [66] Unpentium	344 Uub [65] Unsextium	345 Uut [64] Unseptium	346 Uub [63] Unoctium	347 Uut [62] Unnonium	348 Uuq [61] Undecium	349 Uup [60] Undecium	350 Uub [59] Untrium	351 Uut [58] Untrium	352 Uuq [57] Unquadrium	353 Uup [56] Unpentium	354 Uub [55] Unsextium	355 Uut [54] Unseptium	356 Uub [53] Unoctium	357 Uut [52] Unnonium	358 Uuq [51] Undecium	359 Uup [50] Undecium	360 Uub [49] Untrium	361 Uut [48] Untrium	362 Uuq [47] Unquadrium	363 Uup [46] Unpentium	364 Uub [45] Unsextium	365 Uut [44] Unseptium	366 Uub [43] Unoctium	367 Uut [42] Unnonium	368 Uuq [41] Undecium	369 Uup [40] Undecium	370 Uub [39] Untrium	371 Uut [38] Untrium	372 Uuq [37] Unquadrium	373 Uup [36] Unpentium	374 Uub [35] Unsextium	375 Uut [34] Unseptium	376 Uub [33] Unoctium	377 Uut [32] Unnonium	378 Uuq [31] Undecium	379 Uup [30] Undecium	380 Uub [29] Untrium	381 Uut [28] Untrium	382 Uuq [27] Unquadrium	383 Uup [26] Unpentium	384 Uub [25] Unsextium	385 Uut [24] Unseptium	386 Uub [23] Unoctium	387 Uut [22] Unnonium	388 Uuq [21] Undecium	389 Uup [20] Undecium	390 Uub [19] Untrium	391 Uut [18] Untrium	392 Uuq [17] Unquadrium	393 Uup [16] Unpentium	394 Uub [15] Unsextium	395 Uut [14] Unseptium	396 Uub [13] Unoctium	397 Uut [12] Unnonium	398 Uuq [11] Undecium	399 Uup [10] Undecium	400 Uub [9] Untrium	401 Uut [8] Untrium	402 Uuq [7] Unquadrium	403 Uup [6] Unpentium	404 Uub [5] Unsextium	405 Uut [4] Unseptium	406 Uub [3] Unoctium	407 Uut [2] Unnonium	408 Uuq [1] Undecium	409 Uup [0] Undecium	410 Uub [0] Untrium	411 Uut [0] Untrium	412 Uuq [0] Unquadrium	413 Uup [0] Unpentium	414 Uub [0] Unsextium	415 Uut [0] Unseptium	416 Uub [0] Unoctium	417 Uut [0] Unnonium	418 Uuq [0] Undecium	419 Uup [0] Undecium	420 Uub [0] Untrium	421 Uut [0] Untrium	422 Uuq [0] Unquadrium	423 Uup [0] Unpentium	424 Uub [0] Unsextium	425 Uut [0] Unseptium	426 Uub [0] Unoctium	427 Uut [0] Unnonium	428 Uuq [0] Undecium	429 Uup [0] Undecium	430 Uub [0] Untrium	431 Uut [0] Untrium	432 Uuq [0] Unquadrium	433 Uup [0] Unpentium	434 Uub [0] Unsextium	435 Uut [0] Unseptium	436 Uub [0] Unoctium	437 Uut [0] Unnonium	438 Uuq [0] Undecium	439 Uup [0] Undecium	440 Uub [0] Untrium	441 Uut [0] Untrium	442 Uuq [0] Unquadrium	443 Uup [0] Unpentium	444 Uub [0] Unsextium	445 Uut [0] Unseptium	446 Uub [0] Unoctium	447 Uut [0] Unnonium	448 Uuq [0] Undecium	449 Uup [0] Undecium	450 Uub [0] Untrium	451 Uut [0] Untrium	452 Uuq [0] Unquadrium	453 Uup [0] Unpentium	454 Uub [0] Unsextium	455 Uut [0] Unseptium	456 Uub [0] Unoctium	457 Uut [0] Unnonium	458 Uuq [0] Undecium	459 Uup [0] Undecium	460 Uub [0] Untrium	461 Uut [0] Untrium	462 Uuq [0] Unquadrium	463 Uup [0] Unpentium	464 Uub [0] Unsextium	465 Uut [0] Unseptium	466 Uub [0] Unoctium	467 Uut [0] Unnonium	468 Uuq [0] Undecium	469 Uup [0] Undecium	470 Uub [0] Untrium	471 Uut [0] Untrium	472 Uuq [0] Unquadrium	473 Uup [0] Unpentium	474 Uub [0] Unsextium	475 Uut [0] Unseptium	476 Uub [0] Unoctium	477 Uut [0] Unnonium	478 Uuq [0] Undecium	479 Uup [0] Undecium	480 Uub [0] Untrium	481 Uut [0] Untrium	482 Uuq [0] Unquadrium	483 Uup [0] Unpentium	484 Uub [0] Unsextium	485 Uut [0] Unseptium	486 Uub [0] Unoctium	487 Uut [0] Unnonium	488 Uuq [0] Undecium	489 Uup [0] Undecium	490 Uub [0] Untrium	491 Uut [0] Untrium	492 Uuq [0] Unquadrium	493 Uup [0] Unpentium	494 Uub [0] Unsextium	495 Uut [0] Unseptium	496 Uub [0] Unoctium	497 Uut [0] Unnonium	498 Uuq [0] Undecium	499 Uup [0] Undecium	500 Uub [0] Untrium	501 Uut [0] Untrium	502 Uuq [0] Unquadrium	503 Uup [0] Unpentium	504 Uub [0] Unsextium	505 Uut [0] Unseptium	506 Uub [0] Unoctium	507 Uut [0] Unnonium	508 Uuq [0] Undecium	509 Uup [0] Undecium	510 Uub [0] Untrium	511 Uut [0] Untrium	512 Uuq [0] Unquadrium	513 Uup [0] Unpentium	514 Uub [0] Unsextium	515 Uut [0] Unseptium	516 Uub [0] Unoctium	517 Uut [0] Unnonium	518 Uuq [0] Undecium	519 Uup [0] Undecium	520 Uub [0] Untrium	521 Uut [0] Untrium	522 Uuq [0] Unquadrium	523 Uup [0] Unpentium	524 Uub [0] Unsextium	525 Uut [0] Unseptium	526 Uub [0] Unoctium	527 Uut [0] Unnonium	528 Uuq [0] Undecium	529 Uup [0] Undecium	530 Uub [0] Untrium	531 Uut [0] Untrium	532 Uuq [0] Unquadrium	533 Uup [0] Unpentium	534 Uub [0] Unsextium	535 Uut [0] Unseptium	536 Uub [0] Unoctium	537 Uut [0] Unnonium	538 Uuq [0] Undecium	539 Uup [0] Undecium	540 Uub [0] Untrium	541 Uut [0] Untrium	542 Uuq [0] Unquadrium	543 Uup [0] Unpentium	544 Uub [0] Unsextium	545 Uut [0] Unseptium	546 Uub [0] Unoctium	547 Uut [0] Unnonium	548 Uuq [0] Undecium	549 Uup [0] Undecium	550 Uub [0] Untrium	551 Uut [0] Untrium	552 Uuq [0] Unquadrium	553 Uup [0] Unpentium	554 Uub [0] Unsextium	555 Uut [0] Unseptium	556 Uub [0] Unoctium	557 Uut [0] Unnonium	558 Uuq [0] Undecium	559 Uup [0] Undecium	560 Uub [0] Untrium	561



Chem 11
Final Exam – 2005



ANSWERS

1	A
2	B
3	D

4	C
5	A
6	B

7	A
8	D
9	C

- 10 (a) $n = m \div M = 100 \text{ g} \div 65.02 \text{ g mol}^{-1} = \mathbf{1.54 \text{ mol}}$ (1 mark)
- (b) $\text{moles N}_2 = 1\frac{1}{2} \text{ moles NaN}_3 = 1\frac{1}{2} \times 1.54 \text{ mol} = \mathbf{2.31 \text{ mol}}$ (1 mark)
- $\text{volume N}_2 = n \times 24.79 \text{ L mol}^{-1} = 2.31 \text{ mol} \times 24.79 \text{ L mol}^{-1} = \mathbf{57.2 \text{ L}}$ (1 mark)
- 11 (a) 4 moles initially – 2.3 moles remaining @ Day 16 = **1.7 moles** decomposed (1 mark)
- (b) At **Day 8**, 3 moles of XeF_4 remain and 1 mole of XeF_4 has decomposed yielding 1 mole of Xe and 2 moles of F_2 . (1 mark)
- (c) At Day 40, 3 moles of XeF_4 has decomposed, yielding 6 moles of F_2 .
- $n = m \div M$; $m = n \times M = 6 \text{ mol} \times 38.00 \text{ g mol}^{-1} = \mathbf{228 \text{ g}}$ (1 mark)
- 12 $\text{moles Fe} = m \div M = 2.51 \text{ g} \div 55.85 \text{ g mol}^{-1} = 0.0449 \text{ mol}$
- $\text{moles HCl} = 2 \times \text{moles Fe} = 2 \times 0.0449 \text{ mol} = 0.0899 \text{ mol}$ (1 mark)
- $\text{volume HCl} = n \div c = 0.0899 \text{ mol} \div 1.00 \text{ mol L}^{-1} = \mathbf{0.0899 \text{ L}}$ (1 mark)
- 13 (a) $\text{moles Pb(NO}_3)_2 = \frac{1}{2} \times \text{moles KI} = \frac{1}{2} \times c \times V = \frac{1}{2} \times 0.100 \text{ mol L}^{-1} \times 0.0250 \text{ L}$
- $\text{moles Pb(NO}_3)_2 = \mathbf{0.00125 \text{ mol}}$ (1 mark + 1 mark)
- (b) $\text{moles PbI}_2 = \frac{1}{2} \text{ moles KI} = \frac{1}{2} \times 0.00250 \text{ mol} = 0.00125 \text{ mol}$
- $\text{mass PbI}_2 = n \times M = 0.00125 \text{ mol} \times 461 \text{ g mol}^{-1} = \mathbf{0.576 \text{ g}}$ (1 mark)

- 14 **Sodium thiosulfate was added to hydrochloric acid in a conical flask at various temperatures. The chemicals react to form a precipitate. The time was recorded for how long it took for the precipitate to obscure a cross drawn on the bottom of the conical flask. As the flask was heated, the time taken for the precipitate to obscure the cross became shorter.**

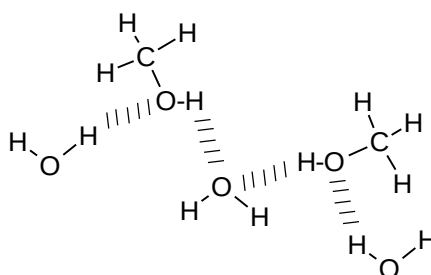
- Detailed description and observed results. 3 marks
 Detailed description of valid experiment. 2 marks
 Simple description of valid experiment. 1 mark

- 15 (a) **Carbon or carbon monoxide or carbon dioxide (1 mark)**

- (b) e.g. $\text{C}_4\text{H}_{10}(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow \text{C}(\text{s}) + \text{CO}(\text{g}) + 2\text{CO}_2(\text{g}) + 5\text{H}_2\text{O}(\text{l})$ (1 mark)

- No states required.

- 16 ||||| indicates hydrogen bonding



Hydrogen bonding shown using dashes or any suitable representation (but not solid lines) between correct atoms. (1 mark)

Written label indicating the hydrogen bond location. (1 mark)

1. Calculate the mass of strontium chloride required to be weighed.

$$\text{Mass of strontium chloride} = 0.250 \text{ L} \times 0.102 \text{ mol L}^{-1} \times [87.62 \times 2(35.45)] \text{ g mol}^{-1} = 4.04 \text{ g}$$

2. Weigh the required quantity and dissolve in the minimum amount of water.
3. Quantitatively transfer the solution to a 250 mL volumetric flask and add enough water until the lower meniscus is just touching the fill line.
4. With the stopper on and held firmly in place, invert the volumetric flask several times to mix the solution.

► Steps 1 and 2 are credited with 1 mark. Numbers 3 and 4 are 1 mark each.

18 (a) mass of iodine = $0.02 \times 250 = 5 \text{ g}$ (1 mark)

(b) mass of ethanol = mass of solution – mass of solute = 245 g (1 mark)

19 (a) Density = mass H_2O $\div$ volume H_2O = $(230.5 \text{ g} - 150.5 \text{ g}) \div 80.3 \text{ mL} = 0.996 \text{ g mL}^{-1}$

- (b) At -10°C , water freezes to a structure where each molecule hydrogen bonds with four other molecules (tetrahedrally configured) creating a regular open structure which occupies more space, hence the greater volume and lower density. (1 mark)

- (c) Ice forms on the surface of a lake (etc.) instead of the bottom allowing life to exist under the ice during the winter.

OR

Ice is an agent of physical weathering for rocks. Successive freezing and thawing can crack open the surface of hard rock forming a component of soil. (1 mark)